

FUNCTIONAL FEATURES GUIDE

Developer: [Left Blank] | OptiCrop Project Documentation
Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Core OptiCrop User Features

- Form Input Validation: Prevents text inputs or negative values in fields like pH and rainfall.
 - Crop Recommendation Display: Renders the recommended crop and system prediction confidence.
 - Yield Forecasting: Displays expected yield in tons per hectare based on seasonal and regional parameters.
 - NPK Diagnostic prescripts: Highlights nutrient levels and provides recommendations for fertilizers like Urea, SSP, and Potash.
 - Visual Analytics: Displays performance charts (confusion matrices, feature rankings) dynamically.
- Opticrop - Functional Features (5 implemented)

1. Crop Recommender

RF classifier recommends optimal crop from 22 types based on N, P, K, pH, rainfall, temperature, humidity.

2. Yield Forecaster

RF regressor estimates harvest yield in t/ha.

3. Soil Health Diagnoser

K-Means (k=5) clusters soil into Excellent / Good / Moderate / Poor / Critical.

4. NPK Prescriber

Computes NPK deficit vs. crop optimum; recommends Urea (N), SSP (P), MOP (K) kg/ha dosage.

5. Analytics Dashboard

Accuracy comparison bar chart, confusion matrix heatmap, feature-importance plot rendered with Matplotlib/Seaborn.